

TitanZero — System Overview

Purpose

TitanZero is an AI-assisted operating system for service businesses.

The platform integrates operations management, communications, automation, marketing, finance, and compliance into a unified environment designed for service industries such as cleaning, property maintenance, and similar field-service businesses.

Rather than operating as a collection of independent tools, TitanZero functions as a coordinated operational intelligence platform where automation, AI reasoning, and business data operate on a shared architecture.

The goal of TitanZero is to create a system where operational signals continuously drive automation, analysis, and intelligent assistance.

Core Philosophy

Traditional SaaS platforms accumulate features over time.

TitanZero is designed around a signal-driven architecture where every meaningful change in the system produces a signal that can trigger automation, AI reasoning, or operational guidance.

The guiding principle of the platform is:

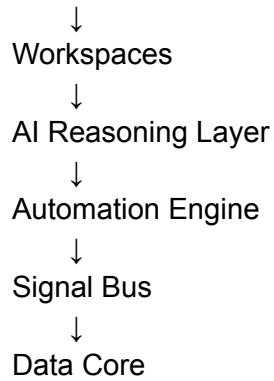
> Every significant event produces a signal, and those signals drive automation and intelligence across the system.

This architecture allows TitanZero to function as a continuously adapting operational system.

System Architecture

TitanZero is organised into a layered architecture.

User Interfaces



At the top of this architecture sits Titan Zero, the system governance intelligence.

Titan Zero — Governance Intelligence

Titan Zero is the overseer of the entire platform.

It functions as the governance layer responsible for:

high-level decision oversight

risk evaluation

planning and system direction

final approval for medium- and high-risk operations

Titan Zero ensures that automation and AI assistants remain aligned with the operational goals and safety constraints of the business.

Most routine actions are handled automatically by the automation and lifecycle assistants.

However, operations that involve financial risk, operational disruption, or significant business decisions require evaluation by Titan Zero.

Titan Zero Cognitive Architecture

Titan Zero operates through a multi-modal open-router intelligence model.

This architecture allows the system to dynamically select the most appropriate AI models for specific reasoning tasks.

The system contains three cooperating reasoning cores.

Titan Zero Core

The Titan Zero Core functions as the central coordination layer.

It manages:

system awareness

signal interpretation

coordination of reasoning processes

integration of outputs from the other cores

OmegaCore

OmegaCore focuses on strategic reasoning and planning.

Responsibilities include:

long-term operational strategy

system optimisation

business growth planning

evaluating large-scale operational decisions

OmegaCore operates at the highest level of abstraction, focusing on the overall direction of the system.

OmicronCore

OmicronCore focuses on fine-grained operational reasoning.

Responsibilities include:

evaluating detailed decisions

reviewing automation actions

analysing operational edge cases

verifying compliance and safety conditions

OmicronCore operates at a granular level, ensuring that decisions are consistent with system rules and operational constraints.

Trinary Bayesian Reasoning Mode

Titan Zero normally operates in standard monitoring mode, where it supervises the system without active intervention.

When a task is classified as medium or high risk, Titan Zero activates Trinary Bayesian Mode.

In this mode:

OmegaCore → strategic evaluation

OmicronCore → detailed operational analysis

Titan Zero Core → synthesis and final decision

The three cores evaluate the decision from different perspectives and combine their reasoning using a Bayesian framework.

This process produces a final approval decision that determines whether the action is permitted, modified, or rejected.

Multi-Modal Open Router

Titan Zero uses a multi-modal open router architecture to access the most appropriate AI models for each reasoning task.

This allows the system to dynamically select:

language reasoning models

planning models

visual or multimodal models

specialist AI models

By routing tasks to the most suitable models, Titan Zero ensures that decisions are based on the best available intelligence.

Data Core

The Data Core stores the factual state of the business.

Operational records include:

customers

jobs

invoices

communications

evidence

automation records

Every operational record belongs to exactly one company.

Tenant ownership is enforced through the `team_id` column, ensuring strict isolation between organisations in the platform.

The Data Core therefore represents the objective truth of the system.

Titan Signal — Event System

Titan Signal is the event bus of the platform.

Whenever an important action occurs, a signal is generated.

Examples include:

job.created
job.completed
invoice.sent
invoice.paid
customer.message.received
campaign.scheduled

Signals are append-only records describing operational events.

They allow the system to react to business activity in real time.

Titan Signal therefore functions as the nervous system of the platform.

Titan Pulse — Automation Engine

Titan Pulse is the automation engine that responds to signals.

Rules follow a simple structure:

Trigger → Condition → Action

Example:

Trigger: job.completed
Condition: job.recurring = true

Action: schedule next visit

Titan Pulse functions as the reflex system of the platform, allowing routine tasks to occur automatically.

Titan Duality — AI Reasoning System

Titan Duality provides AI-assisted operational intelligence.

It consists of two specialised reasoning cores.

LogiCoreAI

LogiCoreAI focuses on operational reasoning.

Responsibilities include:

analysing workflows

detecting risks

suggesting operational improvements

assisting with scheduling and job management

LogiCoreAI manages operations lifecycle assistants.

CreatiCoreAI

CreatiCoreAI focuses on creative and communication intelligence.

Responsibilities include:

marketing campaigns

message generation

promotional strategy

communication drafting

CreatiCoreAI manages marketing lifecycle assistants.

Lifecycle Assistants

Lifecycle assistants are specialised AI agents responsible for guiding business entities through their full lifecycle.

Two primary categories exist.

Operations Lifecycle Assistants

Operations assistants follow operational entities through their lifecycle.

Examples include assistants managing:

quotes

jobs

invoices

customer relationships

Example lifecycle:

Quote

→ Job

→ Completion

→ Invoice

→ Payment

→ Follow-up

These assistants monitor signals and assist with decision making, communication, and automation.

Marketing Lifecycle Assistants

Marketing assistants manage marketing initiatives.

Example lifecycle:

Campaign Created

→ Content Generated

→ Campaign Launched

→ Engagement Monitored

→ Leads Generated

→ Campaign Optimised

These assistants generate creative content and marketing workflows.

Workspaces

Users interact with the system through two primary workspaces.

Logical Workspace

Focused on operational decision making.

Examples include:

dispatch boards

operational inbox

financial alerts

capacity monitoring

Creative Workspace

Focused on building marketing campaigns, automation flows, and communication assets.

User Interfaces

TitanZero provides several interfaces designed for different user preferences.

Titan Command

Titan Command is the primary operational control panel.

It is a single-page command centre that presents the most important operational signals and insights across the business.

Titan Talk

Titan Talk is the communications interface.

It manages:

customer messaging

internal conversations

AI-assisted responses

operational notifications

Titan Origin

Titan Origin provides traditional administrative interfaces.

These include structured screens for managing:

customers

jobs

invoices

system configuration

This interface exists for users who prefer a conventional administrative interface.

Platform Delivery Model

TitanZero is designed as both a desktop web platform and a Progressive Web Application (PWA).

The interface architecture is mobile-first and allows the platform to behave like a native application when installed.

Capabilities include:

installable PWA interface

standalone application mode

responsive design for desktop and mobile

offline-aware functionality

This approach allows TitanZero to operate seamlessly across desktops, tablets, and mobile devices without requiring separate native applications.

Operational Lifecycle

The system follows a continuous service lifecycle.

Lead

- Quote
- Job
- Evidence
- Invoice
- Payment
- Rebooking

Signals generated at each stage enable automation and AI assistance.

Extension Ecosystem

TitanZero supports a modular extension system that allows new capabilities to be added without modifying the core platform.

Extensions follow strict installation and routing rules to ensure stability and tenant isolation.

Summary

TitanZero is a signal-driven business operating system designed to help service companies run their operations through automation, AI assistance, and structured workflows.

The system combines:

a structured data model

a real-time signal bus

an automation engine

lifecycle AI assistants

and a governance intelligence layer

At the centre of the system is Titan Zero, which oversees the platform, evaluates high-risk decisions, and ensures that automation and AI remain aligned with the strategic direction of the business.